

PEP PERFORMANCE REPORT

Flow Analysis | Data Pipeline Methodology

Date:	08 June 2026	Project:	Global Events, Commodities and Currency Markets — DSB Group Project
Prepared By:	Tabassum Nadavinmani (Team Leader) & Abdulhadi Kirkici (Co-Team Leader)	Report Period:	April 2026 – June 2026

1. Summary

This Performance Report summarises the current status, progress, and key performance indicators (KPIs) for the DSB Group Project. The project leverages a Data Pipeline methodology to analyse five datasets: Brent Oil (1946–2026), Gold (1915–2026), Silver (1915–2026), the SWIFT Global Currency Tracker (January–April 2026), and USDG/Global Dollar stablecoin (April 2025–April 2026). The analysis covers the current geopolitical conflict between the United States and Iran, the 2026 Hormuz shipping crisis, energy supply shocks, central bank gold accumulation, and BRICS bloc involvement in non-dollar settlement systems.

As of the report date, all data analytics (Part I - 30 questions + 6 bonus) and market research (Part II -16 questions) work is complete. Charts have been generated and saved. The project is currently in Sprint 3: final report writing, Excel workbook preparation, and submission packaging. The report covers project scope adherence, schedule, resource utilisation, deliverables, risks, and additional relevant areas.

2. Project Scope and Objectives

Scope

Analyse and compare commodity price trends and global currency indicators using a Data Pipeline methodology, connecting observed data patterns to major global events including the Hormuz crisis, energy supply disruptions, gold accumulation, sanctions, BRICS settlement discussions, and currency market shifts.

Datasets in scope:

- Brent Oil price history - 963 monthly observations, 1946–2026
- Gold price history - 1,336 monthly observations, 1915–2026
- Silver price history - 1,336 monthly observations, 1915–2026
- SWIFT Global Currency Tracker - 330 rows, 8 metrics, Jan–Apr 2026
- USDG (Global Dollar stablecoin) - 366 daily observations, Apr 2025–Apr 2026 (collected via CoinGecko API)

Objectives

- Collect and integrate all relevant datasets into a standardised, reproducible pipeline.
- Cleanse, transform, and validate data to ensure analytical integrity.
- Generate descriptive, exploratory, predictive, and prescriptive analytics across four analytical layers.
- Produce comparative visualisations linking commodity and currency movements to global events.
- Deliver actionable, evidence-based insights written for non-technical stakeholders.
- Submit a complete deliverable package report

3. Performance Analysis

A. Work Completed This Period (since the 2026-05-20 report)

- Completed all 16 Part II Market Research questions across sections A-D (Event Identification, Commodity Linking, Currency Linking, Business Interpretation) Abdulhadi Kirkici (technical/data linking) with Parvesh Kumar and Sameep Santosh Dokhale (research).
- Report Writer (Rimsha Shafique) began preliminary review of Part I output and report structure planning ahead of full drafting

B. Work Planned for Next Period

- Write Final Report using Google Docs, incorporating all Part I and Part II findings.
- Build Jupiter Notebook: Raw Data, Cleaned Data, Annual Averages, SWIFT Analysis, Correlation Table.
- Complete all chart insertions into report with proper titles, axis labels, and captions.
- Set up shared Google Drive folder and upload all deliverables in Github.
- Conduct QA review of full report: grammar, consistency, chart labelling, citations.
- Package and submit final deliverable set.

C. Key Accomplishments

- Full analytical coverage: All 30 + 16 + 6 bonus questions answered, no questions skipped or deferred.
- Independent data acquisition: USDG dataset (366 days) collected and integrated via CoinGecko public API with full reproducibility documentation (Python code + citation included).
- Data quality verified across 3,301+ records: 0 critical errors across Brent, Gold, Silver; SWIFT structural NaN confirmed as by-design; 1 USDG duplicate resolved.
- Key market finding — Hormuz crisis: Brent Oil moved from \$102.86/barrel (Jan 2026 baseline) to \$126/barrel peak — a 22% supply-shock spike, the largest short-term movement in the 80-year dataset.
- Key market finding — Gold ATH: Gold reached an intraday all-time high of \$5,595/oz on 29 January 2026, and an annual average of \$4,881.62 in 2026 — a 245× appreciation since 1915.
- Key currency finding: USD retains global payment dominance at 51.14% (April 2026), but its reserve share has declined from 72% to 56.92% over 20 years. CNY rose to SWIFT Rank 5 in April 2026 and BRICS Pay reached \$4B/month operational volume.
- USDG insight: USDG market cap grew from \$226.8M to \$2.42B (~10×) in 12 months — the digital USD layer reinforcing traditional payment dominance in parallel.
- Correlation confirmed: Gold–Silver correlation of 0.95 (annual averages); Brent monthly volatility (9.05%) confirmed as 2.1× higher than Gold (4.30%).

4. Schedule Performance

Milestone	Planned Date	Actual Date	Status	Notes
Project Kickoff	Week of 2026-04-28	2026-04-28	✔ Complete	Assignment analysis + data review initiated
Data Pipeline Design	Week of 2026-05-05	2026-05-07	✔ Complete	Architecture defined; Collect -> Clean -> Transform -> Analyze -> Visualize -> Report
Data Extraction & Cleaning	Week of 2026-05-12	2026-05-15	✔ Complete	All 5 datasets loaded, cleaned, and validated in Jupyter notebook
Preliminary Analysis (Part I)	Week of 2026-05-19	2026-05-19	✔ Complete	All 30 + 6 bonus questions answered; 11 charts generated
In-depth Analysis & Reporting (Part II)	Week of 2026-05-22	2026-05-22	✔ Complete	All 16 market research questions answered
Final Report Submission	2026-07-15	-	● In Progress	Sprint 3 active;

Schedule note: Analysis phases completed on or ahead of plan. Final report phase is on track but the official submission deadline has not yet been confirmed by the course instructor. A clarification request was submitted on 2026-05-18. Sprint 3 is proceeding with an internal target of completing the report draft within 2–3 weeks of this report date.

5. Budget and Resource Utilization

Resource	Planned (hrs)	Actual (hrs)	Variance	Status	%
Data Engineering & Analysis (Abdulhadi Kirkici + Sagar Vig)	30	32	+2	✔ On track	107%
Data Researchers (Parvesh Kumar + Sameep Santosh Dokhale)	20	18	-2	✔ On track	90%
Report Writer (Rimsha Shafique)	15	5	-10	● In progress	33%
Software Tools	0	0	0	✔ All free	—
Total	65	55	-10	Sprint 3 in progress	85%

- Tools used:
- Python 3 + Jupyter Notebook - data pipeline and all analytics
 - pandas, numpy, matplotlib, seaborn, scikit-learn - analysis and visualisation libraries
 - Google Docs - report drafting (collaborative)
 - Trello - task management
 - WhatsApp - team communication

6. Quality and Technical Performance

Quality Indicator	Details
Data Accuracy	All five datasets verified: 0 missing values in Brent/Gold/Silver; SWIFT 388 NaN confirmed as structural (long-format design, not data error — no action taken); 1 USDG duplicate date resolved via drop_duplicates(keep="first"). No negative or zero price values in any commodity dataset.
Pipeline Uptime	100% — the project's analysis notebook executed successfully end-to-end without runtime errors. All 11 charts rendered and saved correctly. A date-parsing issue (MM/DD/YYYY misread as DD/MM/YYYY by default pandas) was identified and resolved using an explicit date format specification.
Compliance	All data handling in line with data source requirements. CoinGecko Public API free tier terms observed: data used for academic research, attribution included, no commercial redistribution. Full data collection methodology and reproducible Python code documented for academic integrity. Citation prepared for all external data sources.

7. Risks and Issues

Risk / Issue	Status	Mitigation Action
Data Gaps — USDG launch period (Nov 2024–Apr 2025) unavailable	Resolved / Accepted	CoinGecko free tier limits historical access to 365 days. First 6 months of USDG history not available. Documented in Data Overview section with explicit limitation note. Sufficient data (366 days) available for all analytical objectives.
Pipeline Integration Bug — Date parsing (MM/DD/YYYY vs DD/MM/YYYY)	Resolved	Explicit format string format='%m/%d/%Y' applied to all commodity date columns. Verified via df.Date.min() / df.Date.max() sanity checks. No data loss.
Team availability variance — Uneven workload distribution	Monitored	Technical workload concentrated on Abdulhadi Kirkici (Co-Leader) as planned. Rimsha Shafique, Sagar Vig, Parvesh Kumar, and Sameep Santosh Dokhale sprint tasks clearly defined for Sprint 3. Capacity monitored via Trello board and async standups.
Final report coordination — Google Drive not yet set up	Open	Tabassum Nadavinmani to create shared Drive folder this week. Report currently being drafted in local Google Docs; will be migrated to shared Drive once folder is available.

8. Variance Analysis

Area	Analysis
Schedule	Analysis phases (Sprint 1 + Sprint 2) completed within the planned timeline. Data pipeline, cleaning, Part I (Q1–Q30 + Bonus), and Part II (Q1–Q16) all delivered. Sprint 3 (final report) is the only phase still in progress. Minor delay risk: official submission deadline not yet received from instructor, preventing a hard end date for Sprint 3. Internal target: report draft complete by 2026-06-22.
Budget (Hours)	Total hours slightly under plan (55 of 65 planned hours used), reflecting that Report Writer (Rimsha Shafique) has not yet begun the report writing phase. Technical work (Abdulhadi Kirkici) ran 2 hours over estimate due to USDG data pipeline complexity and date format debugging. Net variance is within acceptable range; no corrective action required.

9. Forecast

Sprint 3 completion forecast: 2–3 weeks from 2026-06-08, contingent on:

1. Google Drive shared folder setup this week (Tabassum Nadavinmani).
2. Report writing sessions starting this week (Parvesh Kumar + Sameep Santosh Dokhale: Sections 1–3; Sagar Vig: Sections 4–5).

Deliverable readiness forecast:

Deliverable	Forecast Ready Date	Owner	Status
Final Report (Word/PDF)	2026-06-22	Rimsha Shafique (primary), all team review	In Progress
Jupiter Notebook	2026-06-20	Abdulahdi Kirkici	In Progress
PEP Performance Report (signed)	2026-06-15	Abdulahdi Kirkici + Tabassum Nadavinmani	This document
Full Submission Package	2026-06-28	Tabassum Nadavinmani	Not Started

10. Recommendations

1. Prioritise Google Drive setup and report kickoff this week. Rimsha Shafique cannot begin collaborative report drafting until the shared folder is available. This is the single highest-leverage task for the week of 2026-06-08. Target: Drive shared by 2026-06-10.
2. Assign clear section deadlines within Sprint 3. Rimsha Shafique: Sections 1–3 (Introduction, Data Overview, Analytics Findings) by 2026-06-15. Sagar Vig: Section 5 (Final Recommendation) by 2026-06-14 (feeds into Abdulhadi Kirkici's PEP Section 3C). Parvesh Kumar + Sameep Santosh Dokhale: Section 4 (Market Research Findings) by 2026-06-17.
3. Complete QA review before final submission. Abdulhadi Kirkici to review all chart labels, axis titles, and APA references before the report is exported to PDF. All charts must have clear titles and labelled axes per the assignment marking rubric.

Authorizations

Team Leader	Co-Team Leader
Signature: _____ Name: Tabassum Nadavinmani Date: 08 June 2026	Signature: _____ Name: Abdulhadi Kirkici Date: 08 June 2026